

AWS-Native Scalable Disk Monitoring Solution

Multi-Account, Multi-Region Architecture with AWS Organizations

Replacing Ansible while AWS Systems Manager & CloudWatch

Executive Summary

As organizations grow through acquisitions and cloud adoption, managing disk utilization across dozens of AWS accounts spanning multiple regions becomes operationally complex. This document presents a fully AWS-native solution that replaces Ansible-based monitoring with AWS Systems Manager (SSM) and Amazon CloudWatch — governed by AWS Organizations for enterprise-scale automation.

The architecture adopts a hub-and-spoke model: a central monitoring account (designated as Delegated Administrator) receives disk metrics from all spoke accounts and regions via CloudWatch Observability Access Manager (OAM). Service-managed CloudFormation StackSets target Organizational Units (OUs) and automatically deploy the complete monitoring stack to any account joining the Organization — zero manual intervention required.

Service Control Policies (SCPs) at the OU level prevent spoke accounts from disabling or removing monitoring components, ensuring compliance is enforced at the governance layer — not just as a recommendation. EventBridge Organization-wide rules route alarm events from all spoke accounts to a central SNS topic for unified alerting

Problem Statement

Challenge	Detail
Multi-Account Scale	Enterprise grown through acquisitions: multiple AWS accounts under a single AWS Organization, managed centrally via Management Account and OUs
Multi-Region	Accounts span multiple regions (us-east-1, eu-west-1, ap-south-1, etc.); monitoring infra must be deployed per active region
Ansible Dependency	Existing Ansible-based monitoring requires SSH access, bastion hosts, key management, and a dedicated Ansible Tower/AWX server — all being removed in favour of AWS-native services
Dynamic Instances	New EC2 instances launch continuously; monitoring enrollment must be automatic with no manual inventory updates
Governance Gap	Without centralized policy enforcement, spoke accounts can inadvertently disable monitoring. SCPs close this gap at the Organization level.

Solution Architecture

The solution uses an AWS Organizations-governed hub-and-spoke architecture. The Management Account owns the Organization structure and deploys service-managed StackSets. A designated Central Monitoring Account acts as the Delegated Administrator for CloudWatch and SSM, hosting the OAM Sink, CloudWatch Dashboard, SNS topics, and EventBridge rules. Spoke Accounts reside in OUs and receive the complete monitoring stack automatically upon joining the OU.

Architecture Flow Diagram

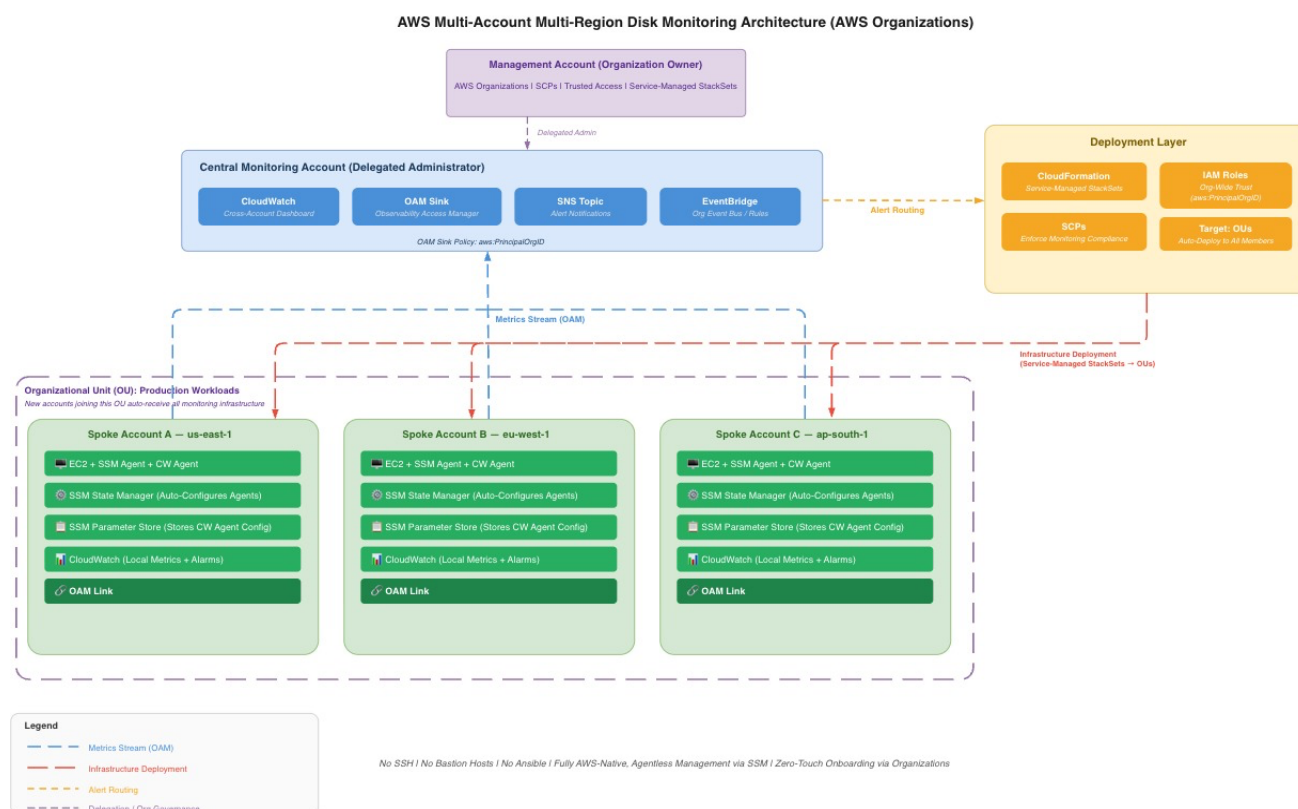


Figure: AWS Organizations Hub-and-Spoke Disk Monitoring Architecture

Key AWS Services

Service	Role	Why Chosen
AWS Organizations	Governs all accounts, OUs, SCPs, and service-managed StackSets	Zero-touch auto-deployment to new accounts
Service-Managed StackSets	Deploy monitoring stack to all accounts in an OU automatically	No manual role setup per account required
AWS Systems Manager	Agent management, remote execution, State Manager associations	Pre-installed on AMIs, no SSH, no bastion
CloudWatch + OAM	Collect disk metrics; stream from all spokes to central sink	Native integration, no Lambda pipelines needed

Service Control Policies	Prevent spoke accounts disabling SSM, CW Agent, or OAM Links	Compliance enforcement at governance layer
EventBridge Org Rules	Route alarm events from all spoke accounts to central SNS	Unified on-call workflow without per-region subscriptions
Delegated Administrator	Central Monitoring Account manages CloudWatch org-wide	Management Account stays clean; ops in dedicated account

AWS Organizations Governance Layer

Trusted Access & Delegated Administrator

Enable trusted access for CloudFormation StackSets and CloudWatch from the Management Account. Designate the Central Monitoring Account as the Delegated Administrator so it can manage cross-account observability without using the Management Account for day-to-day operations.

```
# Run in the MANAGEMENT ACCOUNT
```

```
# Step 1: Enable trusted access for CloudFormation StackSets
```

```
aws organizations enable-aws-service-access \
  --service-principal cloudformation.amazonaws.com
```

```
# Step 2: Enable trusted access for CloudWatch (cross-account observability)
```

```
aws organizations enable-aws-service-access \
  --service-principal cloudwatch.amazonaws.com
```

```
# Step 3: Register Central Monitoring Account as Delegated Administrator
```

```
aws organizations register-delegated-administrator \
  --account-id CENTRAL-MONITORING-ACCOUNT-ID \
  --service-principal cloudformation.amazonaws.com
```

```
aws organizations register-delegated-administrator \
  --account-id CENTRAL-MONITORING-ACCOUNT-ID \
  --service-principal cloudwatch.amazonaws.com
```

Service Control Policies (SCPs) — Enforce Monitoring Compliance

SCPs are attached to the production OU. They prevent spoke accounts from disabling or removing monitoring infrastructure. This ensures compliance is enforced at the governance layer — even if an account administrator attempts to remove monitoring components.

```
// SCP: Deny-DisableMonitoring — attach to Production OU
```

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyDisableSSMAgent",
      "Effect": "Deny",
      "Action": [
        "ssm:DeregisterManagedInstance",
        "ssm:DeleteAssociation"
      ]
    }
  ]
}
```

```

],
"Resource": "*",
"Condition": {
  "ArnNotLike": {
    "aws:PrincipalArn": "arn:aws:iam::*:role/StackSet-DiskMonitoring-*"
  }
}
},
{
  "Sid": "DenyDeleteOAMLink",
  "Effect": "Deny",
  "Action": "oam:DeleteLink",
  "Resource": "*",
  "Condition": {
    "ArnNotLike": {
      "aws:PrincipalArn": "arn:aws:iam::*:role/StackSet-DiskMonitoring-*"
    }
  }
},
{
  "Sid": "DenyModifyMonitoringRole",
  "Effect": "Deny",
  "Action": [
    "iam:DeleteRole",
    "iam:DetachRolePolicy",
    "iam:DeleteRolePolicy"
  ],
  "Resource": "arn:aws:iam::*:role/DiskMonitoringEC2Role",
  "Condition": {
    "ArnNotLike": {
      "aws:PrincipalArn": "arn:aws:iam::*:role/StackSet-DiskMonitoring-*"
    }
  }
}
]
}

```

Component 1: Access & Identity Management

With AWS Organizations, IAM trust no longer requires explicit account IDs. The `aws:PrincipalOrgID` condition key grants trust to any account within the Organization — new accounts are automatically trusted as soon as they join. The SSM Agent communicates via HTTPS outbound (port 443 only), eliminating SSH keys, bastion hosts, and open inbound security group rules.

Ansible vs. SSM: Access Comparison

Aspect	Ansible Approach	SSM Approach (AWS-Native)
Connectivity	SSH/WinRM inbound ports open; bastion hosts per region	HTTPS outbound port 443 only; no inbound ports required

Authentication	SSH key pairs, Ansible Vault, manual rotation	IAM instance profiles — auto-rotated, zero secrets to manage
Multi-Account	Separate inventory and SSH config per account	aws:PrincipalOrgID — single condition covers all Org accounts
New Account	Manual key distribution and inventory registration	Automatic — account joins OU and receives all monitoring infra
Compliance	Manual audit of SSH access logs per region	CloudTrail logs every SSM action; SCPs enforce policy

IAM Trust Policy — Using aws:PrincipalOrgID (Organizations)

The cross-account role uses aws:PrincipalOrgID instead of explicit account IDs. Any account within the Organization can assume this role — no per-account trust policy updates needed when adding new accounts.

```
// IAM Trust Policy — Org-wide cross-account access (no explicit account IDs)
```

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::CENTRAL-MONITORING-ACCOUNT-ID:root"
      },
      "Action": "sts:AssumeRole",
      "Condition": {
        "StringEquals": {
          "aws:PrincipalOrgID": "o-your-org-id"
        }
      }
    }
  ]
}
```

EC2 Instance Profile Policy (SSM + CloudWatch)

```
// EC2 Instance Profile Policy — SSM and CloudWatch permissions
```

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ssm:UpdateInstanceInformation",
        "ssm:ListAssociations",
        "ssm:ListInstanceAssociations",
        "ssmmessages:CreateControlChannel",
        "ssmmessages:CreateDataChannel",
        "ssmmessages:OpenControlChannel",

```

```

    "ssmmessages:OpenDataChannel",
    "ec2messages:GetMessages",
    "ec2messages:AcknowledgeMessage",
    "ec2messages:SendReply"
  ],
  "Resource": "*"
},
{
  "Effect": "Allow",
  "Action": [
    "cloudwatch:PutMetricData",
    "logs:CreateLogGroup",
    "logs:CreateLogStream",
    "logs:PutLogEvents"
  ],
  "Resource": "*"
},
{
  "Effect": "Allow",
  "Action": ["ssm:GetParameter"],
  "Resource": "arn:aws:ssm:*:*:parameter/AmazonCloudWatch-*"
}
]
}

```

Component 2: Data Collection & Aggregation

The CloudWatch Agent is deployed and managed via SSM. It collects disk metrics every 60 seconds and publishes them to a custom namespace `DiskMonitoring/Custom`. Dimensions include `InstanceId`, `AccountId`, `region`, `mount path`, `device`, and `filesystem type` — enabling granular cross-account, cross-region filtering in dashboards.

CloudWatch Agent Configuration (JSON)

This configuration is stored centrally in SSM Parameter Store as `AmazonCloudWatch-DiskMonitoring` in each active region. The agent pulls the config on startup and whenever SSM State Manager re-applies the association.

```

// CloudWatch Agent Configuration file — cw-agent-config.json
{
  "agent": {
    "metrics_collection_interval": 60,
    "run_as_user": "root"
  },
  "metrics": {
    "namespace": "DiskMonitoring/Custom",
    "append_dimensions": {
      "InstanceId": "${aws:InstanceId}",
      "AutoScalingGroupName": "${aws:AutoScalingGroupName}"
    },
    "metrics_collected": {

```

```

"disk": {
  "measurement": [
    "used_percent",
    "free",
    "total",
    "inodes_free"
  ],
  "metrics_collection_interval": 60,
  "resources": ["*"],
  "ignore_file_system_types": ["sysfs", "devtmpfs", "tmpfs", "overlay"]
}
}
}
}

```

Store Config in SSM Parameter Store (CLI)

```

# Bash — Store CloudWatch Agent config in SSM Parameter Store (run per active region)
aws ssm put-parameter \
  --name "AmazonCloudWatch-DiskMonitoring" \
  --type "String" \
  --value file://cw-agent-config.json \
  --description "CloudWatch Agent config for disk utilization monitoring" \
  --tier "Standard" \
  --region <SPOKE_REGION>

```

Component 3: Scalability — Service-Managed StackSets

Service-managed StackSets target Organizational Units (OUs) rather than explicit account IDs. When automatic deployments are enabled, any account joining the OU automatically receives the complete monitoring stack — no manual action, no role creation, no inventory updates.

Zero-Touch Onboarding

Account joins Production OU → StackSet auto-deploys IAM role, instance profile, SSM Parameter Store config, CloudWatch Agent associations, alarms, SNS topic, and OAM Link — all without any manual intervention. New EC2 instances in that account are automatically enrolled by SSM State Manager (targets InstanceIds: "*").

Service-Managed StackSet CloudFormation Template

Deploy this template as a service-managed StackSet from the Management Account (or Delegated Administrator account), targeting the Production OU with automatic deployments enabled. It will deploy to all current member accounts and any future accounts that join the OU.

```

# YAML CloudFormation — Service-Managed StackSet (deploy from Management Account)
AWSTemplateFormatVersion: "2010-09-09"
Description: >
  Disk monitoring infrastructure — deployed via service-managed StackSets
  to all accounts in the Production OU. Auto-deploys to new accounts on OU join.

Resources:
  EC2MonitoringRole:
    Type: AWS::IAM::Role

```

Properties:

RoleName: DiskMonitoringEC2Role

AssumeRolePolicyDocument:

Version: "2012-10-17"

Statement:

- Effect: Allow

Principal:

Service: ec2.amazonaws.com

Action: sts:AssumeRole

ManagedPolicyArns:

- arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
- arn:aws:iam::aws:policy/CloudWatchAgentServerPolicy

EC2InstanceProfile:

Type: AWS::IAM::InstanceProfile

Properties:

InstanceProfileName: DiskMonitoringInstanceProfile

Roles:

- !Ref EC2MonitoringRole

OAM Link — connects spoke account metrics to the central OAM Sink

OAMLink:

Type: AWS::Oam::Link

Properties:

LabelTemplate: \$AccountName

ResourceTypes:

- AWS::CloudWatch::Metric
- AWS::Logs::LogGroup

SinkIdentifier: !Sub >

arn:aws:oam:\${AWS::Region}:CENTRAL-ACCOUNT-ID:sink/SINK-ID

DiskUtilizationAlarm:

Type: AWS::CloudWatch::Alarm

Properties:

AlarmName: HighDiskUtilization-Warning

AlarmDescription: "Disk utilization exceeds 85% - Warning"

Namespace: DiskMonitoring/Custom

MetricName: disk_used_percent

Statistic: Maximum

Period: 300

EvaluationPeriods: 2

Threshold: 85

ComparisonOperator: GreaterThanThreshold

TreatMissingData: breaching

AlarmActions:

- !Ref DiskAlertTopic

DiskUtilizationCritical:

Type: AWS::CloudWatch::Alarm

Properties:

AlarmName: HighDiskUtilization-Critical

AlarmDescription: "CRITICAL - Disk utilization exceeds 95%"


```
Namespace: DiskMonitoring/Custom
MetricName: disk_used_percent
Statistic: Maximum
Period: 60
EvaluationPeriods: 1
Threshold: 95
ComparisonOperator: GreaterThanThreshold
TreatMissingData: breaching
AlarmActions:
  - !Ref DiskAlertTopic
```

```
DiskAlertTopic:
Type: AWS::SNS::Topic
Properties:
  TopicName: DiskUtilizationAlerts
  DisplayName: Disk Space Alerts
```

```
DiskAlertSubscription:
Type: AWS::SNS::Subscription
Properties:
  TopicArn: !Ref DiskAlertTopic
  Protocol: email
  Endpoint: ops-team@yourcompany.com
```

SSM State Manager Associations (CloudFormation YAML)

These two associations are deployed into every spoke account and region via the StackSet. The first installs the CloudWatch Agent package; the second applies the disk monitoring configuration from Parameter Store. By targeting InstanceIds: "*", they catch all current and future EC2 instances automatically.

YAML — SSM State Manager Associations (also included in StackSet template)

Resources:

```
InstallCloudWatchAgent:
Type: AWS::SSM::Association
Properties:
  AssociationName: DeployCloudWatchAgentForDiskMonitoring
  Name: AWS-ConfigureAWSPackage
  Targets:
    - Key: InstanceIds
      Values:
        - "*"
  Parameters:
    action:
      - Install
    name:
      - AmazonCloudWatchAgent
  ScheduleExpression: "rate(1 day)"
  ComplianceSeverity: HIGH
```

```
ConfigureCloudWatchAgent:
Type: AWS::SSM::Association
Properties:
  AssociationName: ConfigureCloudWatchAgentDiskMetrics
```

Name: AmazonCloudWatch-ManagedAgent

Targets:

- Key: InstanceIds

Values:

- "*"

Parameters:

action:

- configure

mode:

- ec2

optionalConfigurationSource:

- ssm

optionalConfigurationLocation:

- AmazonCloudWatch-DiskMonitoring

optionalRestart:

- "yes"

ScheduleExpression: "rate(1 day)"

ComplianceSeverity: HIGH

Scaling: Organizations vs. Self-Managed

Growth Scenario	With AWS Organizations (This Solution)
New account added	Move account into Production OU. StackSet auto-deploys all infra within minutes. Zero manual action.
New region in existing account	Add region to StackSet deployment targets. SSM associations and CW Agent config redeploy automatically.
New EC2 instance	No action needed. SSM State Manager targets InstanceIds: "*" and enrolls new instances on schedule (rate: 1 day).
Config change	Update SSM Parameter Store value. State Manager re-applies the new config across all instances in all accounts on next run.
Org OU restructure	Moving account between OUs automatically adjusts SCP application and StackSet targeting based on OU membership.

Component 4: Centralized Dashboard & Observability (OAM)

What is CloudWatch OAM?

OAM Component	Where It Lives	Purpose
Sink	Central Monitoring Account	Destination that receives all disk metrics from all spoke accounts/regions via OAM Links

Sink Policy	Central Monitoring Account	Uses aws:PrincipalOrgID — all Org accounts trusted automatically, no explicit account list
Link	Each spoke account x each active region	Connects that account+region CloudWatch data to the central sink. Deployed via StackSets.

OAM Setup with Organization-Wide Policy (CLI)

The sink policy uses aws:PrincipalOrgID instead of explicit account IDs. This means all current and future Organization accounts are automatically permitted to create links — the policy never needs to be updated when accounts are added.

Step 1: Create an OAM Sink in the central monitoring account

```
aws oam create-sink \
  --name "DiskMonitoringCentralSink" \
  --tags Key=Purpose,Value=DiskMonitoring
```

Step 2: Create sink policy using aws:PrincipalOrgID (Organization-wide)

This allows ANY account in the Org to create OAM Links — no explicit account IDs needed

```
aws oam put-sink-policy \
  --sink-identifier "arn:aws:oam:us-east-1:CENTRAL-ACCOUNT-ID:sink/SINK-ID" \
  --policy '{
    "Version": "2012-10-17",
    "Statement": [{
      "Effect": "Allow",
      "Principal": "*",
      "Action": ["oam:CreateLink", "oam:UpdateLink"],
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "aws:PrincipalOrgID": "o-your-org-id"
        }
      },
      "ForAllValues:StringEquals": {
        "oam:ResourceTypes": [
          "AWS::CloudWatch::Metric",
          "AWS::Logs::LogGroup"
        ]
      }
    }]
  }'
```

Step 3: In each spoke account+region, create link (deployed via StackSets — see CFn template)

```
aws oam create-link \
  --label-template '$AccountName' \
  --resource-types "AWS::CloudWatch::Metric" \
  --sink-identifier "arn:aws:oam:us-east-1:CENTRAL-ACCOUNT-ID:sink/SINK-ID" \
  --region <SPOKE_REGION>
```

Dashboard Capabilities

- Disk utilization heatmap across all accounts and regions (filterable by Account, Region, Instance, Mount Point)
- Tag-based filtering using common instance tags shared across all Org accounts
- Anomaly detection widgets for trending disk growth (ML-based, no manual threshold tuning)
- Top-N instances by disk utilization ranked across all accounts and all regions
- Alarm state overview — see all WARNING/CRITICAL alarms at a glance from the central account

Component 5: Alerting & EventBridge Organization Rules

CloudWatch Alarms are deployed to every spoke account and region via StackSets. EventBridge Organization-wide rules route alarm events from all spoke accounts to the central SNS topic — enabling a unified on-call workflow without requiring per-account or per-region subscriptions.

Alarm	Threshold	Evaluation	Trigger Condition	Action
Warning	> 85%	2 x 5-min periods	Disk filling — cleanup needed in 24-48h	SNS email to ops-team
Critical	> 95%	1 x 1-min period	Imminent disk full — immediate action required	SNS + optional PagerDuty/Slack
Missing Data	No data	Immediate	Agent stopped reporting — instance or agent issue	SNS alert (breaching state)

EventBridge Organization Rule (Cross-Account Alert Routing)

An EventBridge Organization rule deployed from the Management Account routes CloudWatch Alarm state changes from ALL spoke accounts to the central SNS topic. No per-account EventBridge configuration needed.

```
// EventBridge Organization Rule — deploy from Management Account
{
  "Name": "DiskMonitoringOrgAlarmRoute",
  "EventPattern": {
    "source": ["aws.cloudwatch"],
    "detail-type": ["CloudWatch Alarm State Change"],
    "detail": {
      "alarmName": [{"prefix": "HighDiskUtilization"}],
      "state": {
        "value": ["ALARM"]
      }
    }
  },
  "Targets": [
    {
      "Id": "CentralSNSTopic",
      "Arn": "arn:aws:sns:us-east-1:CENTRAL-ACCOUNT-ID:DiskUtilizationAlerts",
      "RoleArn": "arn:aws:iam::CENTRAL-ACCOUNT-ID:role/EventBridgeOrgAlertRole"
    }
  ]
}
```

```

}
]
}

# CLI: Create Organization-wide event bus rule
aws events put-rule \
  --name "DiskMonitoringOrgAlarmRoute" \
  --event-pattern file://alarm-pattern.json \
  --state ENABLED \
  --event-bus-name "arn:aws:events:us-east-1:MANAGEMENT-ACCOUNT-ID:event-bus/default"

# Enable EventBridge cross-account resource policy on central bus
aws events put-permission \
  --action events:PutEvents \
  --principal "*" \
  --condition '{"Type":"StringEquals","Key":"aws:PrincipalOrgID","Value":"o-your-org-id"}' \
  --statement-id AllowOrgAccounts

```

Advantages of AWS-Native Services Over Ansible

Replacing Ansible with AWS Systems Manager, CloudWatch, and AWS Organizations eliminates multiple layers of operational overhead while delivering superior security, scalability, and compliance enforcement.

Area	Ansible	AWS-Native (Org + SSM + CloudWatch)
Connectivity	SSH/WinRM inbound ports; bastion hosts per region; open SGs	HTTPS outbound port 443 only; no inbound ports; no bastion
Credentials	SSH key pairs, Ansible Vault, manual rotation per account	IAM roles with Org-wide trust — auto-rotated, zero secrets
Inventory	Static host files or dynamic scripts per account/region	Auto-discovery via SSM Fleet Manager (per region, real-time)
New Account	Manual inventory, SSH key distribution per new account	Join OU → zero-touch auto-deploy via service-managed StackSets
Multi-Region	Separate inventory per region; complex jump host routing	Regional SSM + service-managed StackSets deploy natively
High Availability	Dedicated Ansible Tower/AWX EC2 — HA = complex & costly	Fully managed, serverless — AWS manages availability per region
Drift Detection	Manual playbook re-runs on a schedule; no feedback loop	State Manager continuously enforces desired state & reports compliance
Compliance	No guardrails — admins can remove monitoring at will	SCPs prevent disabling monitoring at governance layer

Security Posture	Large attack surface: SSH, bastion, keys per region	Minimal attack surface: IAM + HTTPS + SCP guardrails
Audit Trail	Manual aggregation of SSH logs across accounts/regions	CloudTrail logs every SSM action per region with IAM identity
Visibility	Custom aggregation stack (Elasticsearch/Grafana) required	Native OAM — metrics flow to central account; zero config
Cost	Tower license + dedicated EC2 + operational overhead	SSM free; CW Agent free; minimal metric/alarm charges (~\$200-300/mo)

Key Differentiators:

1. **Zero-Touch Onboarding with Organizations:** New accounts receive full monitoring infrastructure simply by joining the Production OU. StackSets deploy within minutes automatically — no IAM role creation, no inventory updates, no playbook runs.
2. **SCP-Enforced Compliance:** Service Control Policies at the OU level prevent spoke accounts from disabling SSM, removing OAM Links, or deleting monitoring IAM roles — even with Administrator access. Compliance is governance-enforced.
3. **No SSH, No Bastion Hosts:** SSM uses HTTPS outbound via the AWS backbone. Eliminates key rotation, bastion host costs, and open inbound security group rules across every account and region.
4. **No Dedicated Servers:** Ansible Tower/AWX requires a running EC2 with HA configuration per region. SSM is fully managed — no infrastructure to operate, patch, or monitor.
5. **Self-Healing by Design:** SSM State Manager re-applies desired configuration if drift occurs in any account or region, on schedule, automatically. The SCPs prevent drift from being permanent.
6. **Organization-Wide OAM Policy:** The sink policy uses aws:PrincipalOrgID. All Org accounts can link without any policy updates when accounts are added — truly zero-config observability scaling.
7. **Delegated Administrator Pattern:** The Management Account stays clean (no operational workloads). The Central Monitoring Account is the Delegated Administrator, enabling cross-account CloudWatch management without Management Account credentials.
8. **EventBridge Organization Rules:** A single EventBridge rule in the Management Account routes alarm events from ALL member accounts to the central SNS topic — no per-account or per-region configuration.

Cost Analysis

The AWS-native Organizations solution is significantly more cost-efficient than Ansible Tower/AWX. Most core components are either free or have negligible per-unit pricing. Cost scales linearly with the number of instances — NOT with the number of accounts or regions added.

Component	Monthly Cost	Notes
SSM Agent	Free	Pre-installed on Amazon Linux 2, Amazon Linux 2023, Windows AMIs
SSM State Manager	Free	Free for AWS-managed documents (AWS-ConfigureAWSPackage, etc.)

CloudWatch Agent	Free	Software only; metric publication charges apply separately
AWS Organizations	Free	No charge for Organizations, SCPs, OUs, StackSets permissions
CloudFormation StackSets	Free	No additional charge for service-managed StackSet operations
CloudWatch Custom Metrics	~\$0.30/metric	~4 metrics/disk x instances x mount points (first 10K metrics)
CloudWatch Alarms	\$0.10/alarm	2 alarms per instance (Warning 85% + Critical 95%)
CloudWatch OAM	Free	No additional charge for metric sharing via OAM Links
SNS Notifications	First 1M free	Email deliveries; HTTP/S \$0.60/million; very low volume
EventBridge (Org rules)	\$1.00/M events	For cross-account alarm routing; alarm events are very low volume

Conclusion

This AWS Organizations-based disk monitoring solution provides the most streamlined, secure, and scalable approach for multi-account, multi-region environments. By combining Organizations governance (SCPs, OUs, service-managed StackSets), SSM (agentless management), CloudWatch OAM (unified observability), and EventBridge Organization rules (centralized alerting), the architecture eliminates Ansible entirely while delivering superior automation.

New accounts receive full monitoring infrastructure — IAM roles, CloudWatch Agent, alarms, SNS topics, and OAM Links — simply by joining the Production OU. No manual intervention, no role creation, no inventory updates. SCPs ensure that monitoring cannot be accidentally or intentionally disabled by spoke account administrators, enforcing compliance at the governance layer.

The centralized CloudWatch dashboard with Organization-wide OAM provides true single-pane-of-glass visibility, with cross-account and cross-region metric queries available natively. EventBridge Organization rules deliver a unified on-call experience without per-account configuration. The total cost for monitoring 500 VMs across 5 accounts and 3 regions is approximately \$200-300/month — a fraction of the Ansible Tower licensing and infrastructure cost it replaces.